



Brief Paper

Disturbance-observer-based event-triggered model predictive control of nonlinear input-affine systems[☆]Yu Yang, Xiuming Yao^{*}, Hongze Xu

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ABSTRACT

This paper proposes a novel disturbance-observer-based event-triggered model predictive control (DEMPC) framework for a class of nonlinear input-affine systems with state and control input constraints as well as unmatched disturbances to simultaneously enhance the robustness of standard MPC and reduce computational resource utilization. First, a nonlinear disturbance observer is designed to compensate for disturbances actively. Next, a constant-threshold-type event-triggering mechanism (CTETM) is designed in terms of the state prediction deviation caused by the remaining disturbances. Subsequently, a DEMPC algorithm is constructed using the disturbance estimation information, the CTETM, and the so-called dual-mode scheme. Furthermore, rigorous theoretical analysis is provided, involving robust constraint fulfillment, Zeno phenomenon prevention, recursive feasibility, and stability. Particularly for systems with only matched disturbances, it is proven that continuous disturbance compensation would result in a possibly lower triggering frequency and enhanced control performance for the DEMPC compared to conventional EMPC without introducing additional conservatism into the state constraints. In the end, simulation studies are performed to illustrate the effectiveness of the proposed framework.

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1. Introduction

Model predictive control (MPC), as an advanced optimal control approach, has gained tremendous attention concurrently in academia and industry since the mid-20th century due to its capability to handle multivariate constrained systems flexibly and efficiently (Mayne, 2014). The essential idea of MPC is to calculate an optimal control input sequence through the resolution of a well-defined constrained optimization problem at each time instant and then apply the first control action within this sequence as the current input to the plant. This iterative process repeats at subsequent time instants, ensuring the optimal execution of the task (Mayne, Rawlings, Rao, & Scokaert, 2000).

Inevitably, disturbances exist inherently in the majority of industrial control systems, resulting in performance degradation and even system instability. Over the past few decades, in addition to the exploration of the inherent robustness of standard MPC against disturbances, several classical robust MPC approaches have been developed to enhance its robustness, such as

min-max MPC (Limon, Alamo, Salas, & Camacho, 2006), constraint-tightening MPC (Chisci, Rossiter, & Zappa, 2001), and tube-based MPC (Farina & Scattolini, 2012). Furthermore, in contemporary research, disturbance-observer-based MPC (DMPC) has been introduced to enhance the robustness of MPC and has found successful applications in various fields, such as intelligent robots (Sun, Xia, Dai, Liu, & Ma, 2017), aviation (Xu, Yuan, & Zhou, 2021), chemical industries (Kong & Yuan, 2019), etc. Recently, Xie, Dai, Lu, and Xia (2022) has presented a unified DMPC framework along with the corresponding theoretical results for constrained nonlinear input-affine systems with unmatched disturbances.

Another growing research endeavor in MPC involves the development of event-triggered MPC (EMPC) methods with aperiodic implementation. These methods could save system resources while upholding the desired control performance, which holds practical significance, especially for networked control systems with relatively limited communication and computational resources. During the past decade, various event-triggering mechanisms (ETMs) have been investigated, according to a recent survey (Shi & Zhang, 2021). Among these ETMs, the constant-threshold-type ETMs (CTETMs) have been extensively studied in the literature (Li & Shi, 2014; Liu, Gao, Li, & Xu, 2018; Sun, Dai, Xia, & Liu, 2018; Wang, Sun, & Chen, 2019). Although both communication and computational burden could be reduced remarkably by these EMPC approaches, sacrificing the optimal control performance might be unavoidable. The *ad hoc* parameter

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adjustment is required in the design of the above EMPC approaches to achieve the desired trade-off between control performance, and communication and computational burden (Shi & Zhang, 2021). Recently, an appealing effort to simultaneously improve the control performance of EMPC and lower its triggering frequency is to use disturbance rejection techniques. The state prediction deviation could be reduced by active disturbance compensation, thereby potentially lowering the triggering frequency. In Li, Kang, Wang, and Zhao (2022), a disturbance-prediction-based EMPC approach was developed for a class of nonlinear systems with slowly varying disturbances. In Incremona, Ferrara, and Magni (2017), an integral sliding-mode-based EMPC (IEMPC) strategy was proposed for a class of input-affine nonlinear systems with bounded general disturbances. However, the inevitable chattering in integral sliding-mode control would limit its applicability. To provide smooth control input actions, Feng, Wu, Yu, Yamashita, and Huang (2022) investigate a disturbance-observer-based EMPC (DEMPC) scheme for the linearized system subject to control input constraints and matched disturbances.

In this paper, we are particularly interested in developing a novel DEMPC framework for a class of nonlinear input-affine continuous-time systems with unmatched disturbances, and state and control input constraints, to attain a more favorable trade-off between control performance and computational resources. The contributions are summarized as follows:

1. A CTETM is designed based on the state prediction deviation resulting from the remaining unmatched disturbances. Then a DEMPC algorithm is constructed using the disturbance estimation information, the CTETM, and the so-called dual-mode scheme, which would be expected to exhibit enhanced robustness compared to conventional EMPC algorithms while also offering computational resource savings when compared to conventional DMPC algorithms.
2. The DEMPC algorithm applied to a system with only matched disturbances is proved to be equivalent to the corresponding EMPC algorithm applied to a system with gradually alleviating disturbances, albeit with a partial sacrifice in control effort, which does not introduce additional conservatism to state constraints. The triggering frequency of the DEMPC algorithm would be lower compared to the EMPC algorithm while ensuring enhanced robustness.
3. The comprehensive theoretical properties of the proposed DEMPC algorithm, accounting for disturbances and encompassing state and control input constraints, are thoroughly examined. This analysis includes robust constraint fulfillment, Zeno phenomenon prevention, recursive feasibility, and stability. Furthermore, simulation studies are carried out to validate its effectiveness.

The subsequent sections of the article are structured as follows. Section 2 describes the controlled system. Sections 3 and 4 present the DEMPC framework and the main theoretical properties, respectively. Section 5 provides the simulation results. Section 6 finally concludes this paper.

Notations. The set of all non-negative integers and the n -dimensional real Euclidean space are represented by $\mathbf{I}$ and $\mathbf{R}^n$, respectively. $\mathbf{I}(\mathbf{R})_{[a;b]}$ represents $\{x \in \mathbf{I}(\mathbf{R}) \mid a \leq x < b\}$. Given a column vector e and a positive definite matrix $P > 0$, $\gamma_P(e) \triangleq \|e\|_P^2$ and $\gamma_P^{\frac{1}{2}}(e) \triangleq \|e\|_P$ are the P -weighted cost and P -weighted norm, respectively. $\bar{\lambda}(P)(\underline{\lambda}(P))$ is the maximum(minimum) eigenvalue of the matrix P . $\mathcal{B}(i) = \{e \in \mathbf{R}^n \mid \|e\|^2 \leq i^2\}$ denotes the ball of radius i . Furthermore $\sup_{e \in \mathcal{X}} \|e\|$ is the upper bound of $\|e\|$ for $e \in \mathcal{X}$, where $\mathcal{X}$ is a compact set. Given two sets $\mathcal{M}_1, \mathcal{M}_2 \in \mathbf{R}^n$, $\mathcal{M}_1 \ominus \mathcal{M}_2 \triangleq \{z \in \mathbf{R}^n \mid z + \sigma \in \mathcal{M}_1, \forall \sigma \in \mathcal{M}_2\}$ is the Minkowski difference set.

2. System description

A perturbed nonlinear system is considered as:

$$\dot{x}(t) = \mathcal{F}(x(t), u(t)) + \mathcal{D}p(t), t \in \mathbf{R}_{[0;\infty)}, x(0) = x_0, \quad (1)$$

where $\mathcal{F}(x(t), u(t)) \triangleq \mathcal{G}(x(t)) + Bu(t)$, $x(t) \in \mathbf{R}^{N_x}$, $u(t) \in \mathbf{R}^{N_u}$, and $p(t) \in \mathbf{R}^{N_p}$ stand for the system state, control input, and disturbances, respectively. $N_x, N_u, N_p > 0$ are the known constant integers.

The standard assumptions for (1) are stated as follows.

Assumption 1. The matrix $B \in \mathbf{R}^{N_x \times N_u}$ is of full column rank, i.e., $\text{rank}(B) = N_u$.

Assumption 2. The system (1) is assumed to satisfy the state and control input constraints that $x(t) \in \mathcal{X}, u(t) \in \mathcal{U}$, where $\mathcal{X} \subseteq \mathbf{R}^{N_x}$ and $\mathcal{U} \subseteq \mathbf{R}^{N_u}$ are the convex and compact sets with $0 \in \mathcal{X}$ and $0 \in \mathcal{U}$.

Assumption 3. $\mathcal{G}(\cdot) : \mathbf{R}^{N_x} \rightarrow \mathbf{R}^{N_x}$ is the continuously differentiable nonlinear function with $\mathcal{G}(0) = 0$. There exists a Lipschitz constant $0 < \ell < \infty$ such that $\|\mathcal{G}(x_1) - \mathcal{G}(x_2)\| \leq \ell \|x_1 - x_2\|$, for $\forall x_1, x_2 \in \mathcal{X}$.

Assumption 4. The disturbances $p(t)$ are differentiable and satisfy $p(t) \in \mathcal{P} \triangleq \{p(t) \in \mathbf{R}^{N_p} \mid \|p(t)\| \leq \rho_1, \|\dot{p}(t)\| \leq \rho_2\}$, for some constants $\rho_1, \rho_2 > 0$.

Assumption 5. The matrix pair (A, B) is stabilizable where $A \triangleq \frac{\partial \mathcal{G}(x)}{\partial x}|_{x=0} \in \mathbf{R}^{N_x \times N_x}$, that is, $A_k \triangleq A + BK$ is Hurwitz for a linear state-feedback gain $K \in \mathbf{R}^{N_u \times N_x}$.

In accordance with Assumption 5, we present a well-established result as follows.

Lemma 1 (Chen & Allgower, 1998). For the system $\dot{x}(t) = \mathcal{F}(x(t), u(t))$, suppose that Assumption 5 holds. Given two matrixes $Q, R > 0$, then there exist a constant $\gamma > 0$ and a matrix $P > 0$ such that: 1) $Kx(t) \in \mathcal{U}$, for $\forall x(t) \in \Omega(\gamma) \triangleq \{x \in \mathbf{R}^{N_x} \mid \gamma_P(x) \leq \gamma^2\}$; 2) $\Omega(\gamma)$ is invariant for the closed-loop system $\dot{x}(t) = \mathcal{F}(x(t), Kx(t))$; 3) $\dot{\gamma}_P(x(t)) \leq -\gamma_{Q^*}(x(t))$, for $\forall x(t) \in \Omega(\gamma)$, where $Q^* \triangleq Q + K^T RK$.

3. Design of DEMPC framework

In this section, the DEMPC framework is developed to robustly stabilize the system (1). The composite control input $u(t)$ consists of two components as follows:

$$u(t) = u_{do}(t) + u_{empc}(t), \quad (2)$$

where $u_{empc}(t), u_{do}(t) \in \mathbf{R}^{N_u}$ are the EMPC input and disturbance compensation input, respectively.

3.1. Design of $u_{do}(t)$

A disturbance observer is constructed to estimate $p(t)$ as

$$\begin{cases} \dot{\hat{\mu}}(t) = -\mathcal{L}[\mathcal{G}(x(t)) + Bu(t) + \mathcal{D}(\mu(t) + \mathcal{L}x(t))] \\ \hat{p}(t) = \mu(t) + \mathcal{L}x(t), \end{cases} \quad (3)$$

where $\hat{p}(t) \in \mathbf{R}^{N_p}$, with $\hat{p}(0) = 0$, is the disturbance estimate. $\mathcal{L} \in \mathbf{R}^{N_p \times N_x}$ and $\mu(t) \in \mathbf{R}^{N_p}$ are the observer gain and auxiliary variable, respectively. The disturbance estimation error is defined as $\tilde{p}(t) \triangleq p(t) - \hat{p}(t)$. The dynamics of $\tilde{p}(t)$ is then obtained that

$$\dot{\tilde{p}}(t) = -\mathcal{L}\mathcal{D}\tilde{p}(t) + \dot{p}(t). \quad (4)$$

Note that the dynamics (4) is input-to-state stable when choosing an appropriate $\mathcal{L}$ such that $-\mathcal{L}\mathcal{D}$ is Hurwitz. Regarding (3) and (4), the following lemma can be used to estimate the upper bounds of $\|\hat{p}(t)\|$ and $\|\tilde{p}(t)\|$.

Lemma 2 (Xie et al., 2022). For the dynamics (4), suppose that Assumption 4 holds. If $-\mathcal{L}\mathcal{D}$, denoted as $\mathcal{E}$, is constant and Hurwitz, it can be shown that

$$\begin{aligned}\|\hat{p}(t)\| &\leq \frac{\rho_1 \|\mathcal{E}\| \beta(\mathcal{E})}{\alpha(\mathcal{E})} (1 - e^{-\alpha(\mathcal{E})t}), \\ \|\tilde{p}(t)\| &\leq \beta(\mathcal{E}) \left(\rho_1 - \frac{\rho_2}{\alpha(\mathcal{E})} \right) e^{-\alpha(\mathcal{E})t} + \frac{\rho_2 \beta(\mathcal{E})}{\alpha(\mathcal{E})},\end{aligned}$$

where $\beta(\mathcal{E}) = \sqrt{\|H\| \|H^{-1}\|}$, $\alpha(\mathcal{E}) = 1/(2\|H\|)$. H is the Hermitian matrix satisfying that $H\mathcal{E} + \mathcal{E}^T H = -I$.

Remark 1. The estimates of the upper bounds of $\|\hat{p}(t)\|$ and $\|\tilde{p}(t)\|$ would rely on the chosen matrix $\mathcal{E}$ (i.e., the observer gain $\mathcal{L}$). A discussion on the estimation precisions of the upper bounds can be found in Xie et al. (2022).

To derive $u_{do}(t)$, the following lemma is recalled.

Lemma 3 (Castanos & Fridman, 2006). For the matrix $B \in \mathbf{R}^{\mathcal{N}_x \times \mathcal{N}_u}$ satisfying Assumption 1, it holds that

$$I = BB^+ + B^\perp B^{\perp+}, \quad (5)$$

where I is the $\mathcal{N}_x$ -dimensional identity matrix. $B^+ = (B^T B)^{-1} B^T$ is the left inverse of B . The column vectors of $B^\perp \in \mathbf{R}^{\mathcal{N}_x \times (\mathcal{N}_x - \mathcal{N}_u)}$ span the null space of B^T .

Substituting (5) into (1) yields that

$$\dot{x}(t) = \mathcal{G}(x(t)) + Bu(t) + p_m(t) + p_{um}(t), \quad (6)$$

where $p_m(t) = BB^+ \mathcal{D}p(t)$ and $p_{um}(t) = B^\perp B^{\perp+} \mathcal{D}p(t)$ are the matched and unmatched disturbances, respectively. Thus, $u_{do}(t)$ is set as

$$u_{do}(t) \triangleq -B^+ \mathcal{D}\hat{p}(t). \quad (7)$$

Substituting (2) and (7) into (6) yields that

$$\dot{x}(t) = \mathcal{G}(x(t)) + Bu_{empc}(t) + \zeta(t), \quad (8)$$

where $\zeta(t) = \mathcal{D}\tilde{p}(t) + B^\perp B^{\perp+} \mathcal{D}\hat{p}(t)$ is defined as the remaining disturbances, with $\zeta_m \triangleq \sup_{t \in \mathbf{R}_{[0;\infty)}} \|\zeta(t)\|$.

3.2. Design of $u_{empc}(t)$

3.2.1. Optimization problem

Consider the nominal formulation of (8) as

$$\dot{\bar{x}}(t) = \mathcal{G}(\bar{x}(t)) + B\bar{u}(t), \quad (9)$$

where $\bar{x}(t) \in \mathbf{R}^{\mathcal{N}_x}$ and $\bar{u}(t) \in \mathbf{R}^{\mathcal{N}_u}$ denote the nominal system state and control input, respectively. Denote the triggering time series as $\{t_k\}$, where $k \in \mathbf{I}_{[0;\infty)}$ and $t_0 = 0$. Then a nonlinear constrained optimization problem is constructed as follows.

Problem 1. $\bar{u}^0(\cdot|t_k) = \arg \min_{\bar{u}(\cdot|t_k) \in \mathcal{U}} \mathcal{J}(\bar{u}(\cdot|t_k), x(t_k))$ subject to

$$\begin{aligned}\bar{x}(t_k|t_k) &= x(t_k) \\ \dot{\bar{x}}(v|t_k) &= \mathcal{G}(\bar{x}(v|t_k)) + B\bar{u}(v|t_k), \quad v \in \mathbf{R}_{[t_k:t_k+\mathcal{T})} \\ \bar{x}(v|t_k) &\in \mathcal{X}(v - t_k) \\ \bar{x}(t_k + \mathcal{T}|t_k) &\in \Omega(\epsilon),\end{aligned}$$

where $\bar{u}(\cdot|t_k) \in \mathbf{R}^{\mathcal{N}_u}$ and $\bar{x}(\cdot|t_k) \in \mathbf{R}^{\mathcal{N}_x}$ represent the feasible control input and the corresponding predicted state trajectories, respectively, at t_k . $\mathcal{T} > 0$ is the fixed prediction horizon. Referring to Chisci et al. (2001), the robust tightened constraint sets are defined as

$$\mathcal{X}(v - t_k) \triangleq \mathcal{X} \ominus \mathcal{B}(\zeta(v - t_k)), \quad (10a)$$

$$\mathcal{U} \triangleq \mathcal{U} \ominus \mathcal{B}(\phi). \quad (10b)$$

Note that $\phi = \|B^+ \mathcal{D}\| \frac{\rho_1 \|\mathcal{E}\| \beta(\mathcal{E})}{\alpha(\mathcal{E})}$ is obtained according to (7) and Lemma 2. $\zeta(s) = \frac{\zeta_m}{\ell} (e^{\ell s} - 1)$ and $\Omega(\epsilon) = \{x \mid \mathcal{V}_P(x) \leq \epsilon^2, \epsilon > 0\}$ are derived according to Theorems 1–3. The cost function $\mathcal{J}(\cdot, \cdot)$ at t_k is designed as follows:

$$\begin{aligned}\mathcal{J}(\bar{u}(\cdot|t_k), x(t_k)) &\triangleq \int_{t_k}^{t_k+\mathcal{T}} \mathcal{E}(\bar{x}(v|t_k), \bar{u}(v|t_k)) dv \\ &\quad + \mathcal{V}_P(\bar{x}(t_k + \mathcal{T}|t_k)),\end{aligned}$$

where $\mathcal{E}(\bar{x}, \bar{u}) \triangleq \mathcal{V}_Q(\bar{x}) + \mathcal{V}_R(\bar{u})$. $Q, R, P > 0$ are the designed weights. Then, the optimal control input trajectory $\bar{u}^0(\cdot|t_k)$ is obtained by solving Problem 1, also the corresponding nominal optimal predicted state trajectory $\bar{x}^0(\cdot|t_k)$. The optimal cost function at t_k is denoted as $\mathcal{J}^0(x(t_k))$. Thus, the EMPC input is set as

$$u_{empc}(t) \triangleq \bar{u}^0(t|t_k), \quad t \in \mathbf{R}_{[t_k:t_{k+1})}.$$

In the sequel, a CTETM will be built based on the state prediction deviation to determine $\{t_k\}$.

3.2.2. CTETM

For $v \in \mathbf{R}_{[t_k:t_{k+1})}$, the state trajectory $x(v)$ of the system (1) under the control input (2) would always deviate from the state trajectory $\bar{x}^0(v|t_k)$ of the system (9) under the control input $\bar{u}^0(v|t_k)$, in the presence of residual disturbances. According to this fact, the triggering condition is designed as

$$t'_{k+1} \triangleq \inf_{v \in \mathbf{R}_{[t_k:t_{k+1})}} \{v : \|x(v) - \bar{x}^0(v|t_k)\| = \varpi\}, \quad (11a)$$

$$t_{k+1} \triangleq \min\{t'_{k+1}, t_k + \mathcal{T}\}, \quad (11b)$$

where $\varpi \triangleq \frac{\zeta_m}{\ell} (e^{\ell \beta \mathcal{T}} - 1)$, with $\beta \in \mathbf{R}_{(0;1)}$, is the triggering threshold. The next theorem is provided to guarantee the Zeno phenomenon prevention of the DEMPC framework.

Theorem 1. For the system (1), suppose that Assumptions 1–4 hold. If the triggering time series $\{t_k\}$ is determined as (11), the lower and upper bounds of the inter-execution time satisfy that $\inf_{k \in \mathbf{I}_{[0;\infty)}} \{t_{k+1} - t_k\} \geq \beta \mathcal{T}$ and $\sup_{k \in \mathbf{I}_{[0;\infty)}} \{t_{k+1} - t_k\} \leq \mathcal{T}$, respectively.

Proof. First, it can be obtained directly from (11b) that $\sup_{k \in \mathbf{I}_{[0;\infty)}} \{t_{k+1} - t_k\} \leq \mathcal{T}$. Second, for $v \in \mathbf{R}_{[t_k:t_{k+1})}$, we have that $\|x(v) - \bar{x}^0(v|t_k)\| \leq \ell \int_{t_k}^v \|x(v) - \bar{x}^0(v|t_k)\| dv + \zeta(t_k, v)(v - t_k) \leq \ell \int_{t_k}^v \|x(v) - \bar{x}^0(v|t_k)\| dv + \zeta_m(v - t_k)$, where $\zeta(t_k, v) \triangleq \sup_{s \in \mathbf{R}_{[t_k:v]}} \|\zeta(s)\|$. Using the Gronwall–Bellman inequality yields that $\|x(v) - \bar{x}^0(v|t_k)\| \leq \frac{\zeta_m}{\ell} (e^{\ell(v-t_k)} - 1)$. From (11a), we have that $t'_{k+1} \geq t_k + \beta \mathcal{T}$. It holds according to (11b) that $\inf_{k \in \mathbf{I}_{[0;\infty)}} \{t_{k+1} - t_k\} \geq \beta \mathcal{T}$. By summarizing these two steps, the proof is finalized.

Remark 2. With reference to Corollary 1 in Xie et al. (2022), when there only exist matched disturbances (i.e., $B = \mathcal{D}$), by choosing an appropriate $\mathcal{L}$, it can be derived that $\zeta(t) = \mathcal{D}\tilde{p}(t)$ and $\zeta_m = \|\mathcal{D}\| \rho_1$. Therefore, the triggering condition (11) designed based on the upper bound of the remaining disturbances will be consistent with the triggering condition designed in Liu et al. (2018) based on the upper bound of the original disturbances and does not require information on the first-order derivative of the disturbances. In addition, the upper bound of the remaining disturbances decreases exponentially over time (Xie et al., 2022). The tightened state constraint (10a) will be the same as that in Liu et al. (2018) due to $\zeta_m = \|\mathcal{D}\| \rho_1$. Therefore, it can be proved that, with the continuous active disturbance compensation as a portion of the control effort in the DEMPC scheme, the

state prediction deviation would be mitigated gradually, which could facilitate not only greater control performance but also lower triggering frequency when compared to the corresponding EMPC scheme.

Remark 3. It is noteworthy that, because of the inherently composite design structure of the DEMPC scheme, it seamlessly integrates with any existing EMPC scheme. This fashion preserves the fundamental principles of the EMPC scheme. In other words, for any established EMPC scheme, a corresponding DEMPC scheme can be readily established. The key distinction lies that the DEMPC scheme compensates for disturbances by sacrificing a portion of the control effort, which introduces a level of conservatism to the tightened constraints to improve control performance while simultaneously reducing computational resource usage.

3.3. DEMPC algorithm

Referring to [Michalska and Mayne \(1993\)](#), a dual-mode based DEMPC algorithm is outlined in Algorithm 1. That is, the locally stabilizing state-feedback controller $u_{empc}(t) \triangleq Kx(t) \in \mathcal{U}$ is enabled if the system state $x(t)$ is inside the set $\Omega(\gamma)$.

Algorithm 1 DEMPC

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1: while  $x(t) \notin \Omega(\gamma)$  do
2:   Solve Problem 1 at  $t_k$ 
3:   while triggering instant  $t_{k+1}$  is not reached do
4:     Apply  $u(t) \leftarrow u_{do}(t) + (u_{empc}(t) \leftarrow \tilde{u}^0(t|t_k))$ 
5:   end while
6:    $k \leftarrow k + 1$ 
7: end while
8: Apply  $u(t) \leftarrow u_{do}(t) + (u_{empc}(t) \leftarrow Kx(t))$ 

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Remark 4. The conservatism of the algorithm would result from the global estimation nature of the constants ζ_m and ℓ . Using the local Lipschitz constant to reduce the conservatism could be found in [Marruedo, Bravo, Alamo, and Camacho \(2002\)](#). In addition, time-varying MPC might be a clue to handle the time-varying parameter $\zeta(t_k, v)$ to further reduce the conservatism.

4. Theoretical analysis

Prior to substantiating the recursive feasibility and stability results, a fundamental assumption is first stated as follows.

Assumption 6. There is at least one feasible solution to [Problem 1](#) at the initial time instant t_0 .

4.1. Recursive feasibility

The following lemma is used in the proof of [Theorem 2](#).

Lemma 4. For $\forall \eta_1, \eta_2 \in \mathbf{R}^{N_x}$ and $v \in \mathbf{R}_{[t_{k+1}:t_k+\mathcal{T}]}$, if the conditions hold that $\|\eta_2 - \eta_1\| \leq \varpi e^{\ell(v-t_{k+1})}$ and $\eta_1 \in \tilde{\mathcal{X}}(v - t_k)$, it can be obtained that $\eta_2 \in \tilde{\mathcal{X}}(v - t_{k+1})$.

Proof. For $\forall s \in \mathcal{B}(\zeta(v - t_{k+1}))$, it is seen that $\|s\| \leq \frac{\zeta_m}{\ell}(e^{\ell(v-t_{k+1})} - 1)$. Define the auxiliary variable $\tau = s + \eta_2 - \eta_1$. Using the triangle inequality, $\varpi \triangleq \frac{\zeta_m}{\ell}(e^{\ell\beta\mathcal{T}} - 1)$, and $t_{k+1} - t_k \geq \beta\mathcal{T}$ yields that $\|\tau\| \leq \frac{\zeta_m}{\ell}(e^{\ell(v-t_{k+1})} - 1) + \varpi e^{\ell(v-t_{k+1})} \leq \frac{\zeta_m}{\ell}(e^{\ell(v-t_k)} - 1)$ thus $\tau \in \mathcal{B}(\zeta(v - t_k))$. Because of $\eta_1 \in \tilde{\mathcal{X}}(t - t_k) \triangleq \mathcal{X} \ominus \mathcal{B}(\zeta(v - t_k))$, it follows the Minkowski set subtraction that $\eta_1 + \tau \in \mathcal{X}$, which implies that $\eta_2 + s \in \mathcal{X}$. Further, due to $s \in \mathcal{B}(\zeta(v - t_{k+1}))$, it holds that $\eta_2 \in \tilde{\mathcal{X}}(v - t_{k+1})$.

Theorem 2. For the system (1), suppose that [Assumptions 1–6](#) hold. Algorithm 1 will be feasible for $t_k, k \in \mathbf{I}_{[1:\infty)}$, if the following conditions are met:

$$\sqrt{\lambda(P)}\varpi e^{\ell\mathcal{T}(1-\beta)} \leq \gamma - \epsilon, \quad (12)$$

$$\mathcal{T} \geq 2 \frac{\bar{\lambda}(P)}{\lambda(Q^*)\beta} \ln \frac{\gamma}{\epsilon}, \quad (13)$$

$$\Omega(\gamma) \subseteq \mathcal{X} \ominus \mathcal{B}(\zeta(\mathcal{T})). \quad (14)$$

Proof. Consider the candidate control input that

$$\tilde{u}(v|t_{k+1}) = \begin{cases} \tilde{u}^0(v|t_k), & v \in \mathbf{R}_{[t_{k+1}:t_k+\mathcal{T}]}, \\ K\tilde{x}(v|t_{k+1}), & v \in \mathbf{R}_{[t_k+\mathcal{T}:t_{k+1}+\mathcal{T}]}, \end{cases} \quad (15)$$

where $\tilde{x}(v|t_{k+1})$ is the predicted state under the control input $\tilde{u}(v|t_{k+1})$, with $\tilde{x}(t_{k+1}|t_{k+1}) = x(t_{k+1})$.

First, we show that $\tilde{x}(v|t_{k+1})$ satisfies the terminal constraint. For $\forall v \in \mathbf{R}_{[t_{k+1}:t_k+\mathcal{T}]}$, by using (11a), it is seen that $\|\tilde{x}(v|t_{k+1}) - \tilde{x}^0(v|t_k)\| \leq \varpi + \ell \int_{t_k}^v \|\tilde{x}(s|t_{k+1}) - \tilde{x}^0(s|t_k)\| ds$. By using the Gronwall–Bellman inequality, it is seen that

$$\|\tilde{x}(v|t_{k+1}) - \tilde{x}^0(v|t_k)\| \leq \varpi e^{\ell(v-t_{k+1})}. \quad (16)$$

Substituting $v = t_k + \mathcal{T}$ into (16) yields that $\|\tilde{x}(t_k + \mathcal{T}|t_{k+1}) - \tilde{x}^0(t_k + \mathcal{T}|t_k)\| \leq \varpi e^{\ell(t_k+\mathcal{T}-t_{k+1})}$. Then, by using the triangle inequality, Rayleigh inequality, and $t_{k+1} - t_k \geq \beta\mathcal{T}$, it follows (12) that $\gamma_P^{\frac{1}{2}}(\tilde{x}(t_k + \mathcal{T}|t_{k+1})) \leq \sqrt{\lambda(P)}\varpi e^{\ell(\mathcal{T}-\beta\mathcal{T})} + \epsilon \leq \gamma$. Further, by using the comparison principle and [Lemma 1](#), it is seen that $\gamma_P(\tilde{x}(v|t_{k+1})) \leq \gamma_P(\tilde{x}(t_k + \mathcal{T}|t_{k+1}))e^{-\frac{\lambda(Q^*)}{\lambda(P)}(v-t_k-\mathcal{T})}$ for $\forall v \in \mathbf{R}_{[t_k+\mathcal{T}:t_{k+1}+\mathcal{T}]}$. Then, it is obtained from (13) that $\gamma_P(\tilde{x}(t_{k+1} + \mathcal{T}|t_{k+1})) \leq \epsilon^2$.

Second, we show that the candidate control input satisfies the tightened constraint $\tilde{\mathcal{U}}$. It can be directly seen from (15) that $\tilde{u}(v|t_{k+1}) \in \tilde{\mathcal{U}}$, for $\forall v \in \mathbf{R}_{[t_{k+1}:t_k+\mathcal{T}]}$. Furthermore, because of $\tilde{x}(t_k + \mathcal{T}|t_{k+1}) \in \Omega(\gamma)$, $K\tilde{x}(v|t_{k+1})$ is enabled. Then, it follows [Lemma 1](#) that for $\forall v \in \mathbf{R}_{[t_k+\mathcal{T}:t_{k+1}+\mathcal{T}]}$, $\tilde{u}(v|t_{k+1}) \in \tilde{\mathcal{U}}$ holds.

Third, we will prove that the state $\tilde{x}(v|t_{k+1})$ satisfies the tightened constraint $\tilde{\mathcal{X}}(v - t_{k+1})$. For $\forall v \in \mathbf{R}_{[t_{k+1}:t_k+\mathcal{T}]}$, applying [Lemma 4](#) into (16) yields that $\tilde{x}(v|t_{k+1}) \in \tilde{\mathcal{X}}(v - t_{k+1})$. Then, for $\forall v \in \mathbf{R}_{[t_k+\mathcal{T}:t_{k+1}+\mathcal{T}]}$, by using (14), it is seen that $\tilde{x}(v|t_{k+1}) \in \Omega(\gamma) \subseteq \mathcal{X} \ominus \mathcal{B}(\zeta(\mathcal{T})) \subseteq \tilde{\mathcal{X}}(v - t_{k+1})$.

Based on the above, it can be seen that (15) is a feasible solution to [Problem 1](#) at t_{k+1} . Thus, the recursive feasibility of Algorithm 1 can be guaranteed.

4.2. Stability

Theorem 3. For the system (1) under the control input (2), with that [Assumptions 1–6](#) and the conditions (12)–(14) are met, the system state will, in finite time, enter the set $\Omega(\gamma)$ if it holds:

$$\begin{aligned} & \bar{\lambda}(Q)(1 - \beta)\mathcal{T}\varpi e^{\ell(\mathcal{T}-\beta\mathcal{T})}(\varpi e^{\ell(\mathcal{T}-\beta\mathcal{T})} + 2\tilde{\psi}) \\ & + \sqrt{\lambda(P)}\varpi e^{\ell(\mathcal{T}-\beta\mathcal{T})}(\gamma + \epsilon) < \beta\mathcal{T}\bar{\lambda}(Q)\left(\frac{\gamma}{\sqrt{\lambda(P)}} - \varpi\right)^2, \end{aligned} \quad (17)$$

where $\tilde{\psi} = \sup_{x \in \mathcal{X}} \|x\| - \zeta(\beta\mathcal{T})$. Afterward, the closed-loop system will remain stable and the system state trajectory will converge to the set $\Omega(\bar{\gamma})$ eventually, if it holds that $\bar{\gamma} \leq \gamma$, where $\bar{\gamma} = \sqrt{\frac{2\bar{\lambda}(P)\|\sqrt{P}\|}{\lambda(Q^*)}}\gamma\tilde{\zeta}_m$.

Proof. The theorem is proved by discussing the two cases of the initial state, i.e., $x(t_k) \notin \Omega(\gamma)$ and $x(t_k) \in \Omega(\gamma)$.

First, for $x(t_k) \notin \Omega(\gamma)$, define the error cost function as

$$\mathcal{J}_{\Delta}^{0,k} = \mathcal{J}^0(x(t_{k+1})) - \mathcal{J}^0(x(t_k)).$$

According to the suboptimality of the candidate control input (15), it can be obtained that

$$\begin{aligned} \mathcal{J}_{\Delta}^{0,k} &\leq \mathcal{J}(\tilde{u}(\cdot|t_{k+1}), x(t_{k+1})) - \mathcal{J}^0(x(t_k)) \\ &= \int_{t_{k+1}}^{t_{k+1}+\mathcal{T}} \mathcal{V}_Q(\tilde{x}(v|t_{k+1})) + \mathcal{V}_R(\tilde{u}(v|t_{k+1})) dv \\ &\quad + \mathcal{V}_P(\tilde{x}(t_{k+1} + \mathcal{T}|t_{k+1})) - \mathcal{V}_P(\tilde{x}(t_k + \mathcal{T}|t_k)) \\ &\quad - \int_{t_k}^{t_k+\mathcal{T}} \mathcal{V}_Q(\bar{x}^0(v|t_k)) + \mathcal{V}_R(\bar{u}^0(v|t_k)) dv. \end{aligned} \quad (18)$$

According to Lemma 1 and (15), it is seen that

$$\begin{aligned} &\mathcal{V}_P(\tilde{x}(t_{k+1} + \mathcal{T}|t_{k+1})) - \mathcal{V}_P(\tilde{x}(t_k + \mathcal{T}|t_k)) \\ &\leq - \int_{t_k+\mathcal{T}}^{t_{k+1}+\mathcal{T}} \mathcal{V}_Q(\tilde{x}(v|t_{k+1})) + \mathcal{V}_R(\tilde{u}(v|t_{k+1})) dv. \end{aligned} \quad (19)$$

Substituting (15) and (19) into (18) yields that

$$\mathcal{J}_{\Delta}^{0,k} \leq \sum_{i=1}^3 \mathcal{J}_{\Delta i}^{0,k},$$

where

$$\mathcal{J}_{\Delta 1}^{0,k} = \int_{t_{k+1}}^{t_{k+1}+\mathcal{T}} \mathcal{V}_Q(\tilde{x}(v|t_{k+1})) - \mathcal{V}_Q(\bar{x}^0(v|t_k)) dv, \quad (20a)$$

$$\mathcal{J}_{\Delta 2}^{0,k} = \mathcal{V}_P(\tilde{x}(t_k + \mathcal{T}|t_{k+1})) - \mathcal{V}_P(\bar{x}^0(t_k + \mathcal{T}|t_k)), \quad (20b)$$

$$\mathcal{J}_{\Delta 3}^{0,k} = - \int_{t_k}^{t_k+\mathcal{T}} \mathcal{V}_Q(\bar{x}^0(v|t_k)) + \mathcal{V}_R(\bar{u}^0(v|t_k)) dv. \quad (20c)$$

For (20a), by using Rayleigh inequality, and Holder inequality, it is observed that

$$\begin{aligned} \mathcal{J}_{\Delta 1}^{0,k} &\leq 2\bar{\lambda}(Q) \int_{t_{k+1}}^{t_{k+1}+\mathcal{T}} \|\tilde{x}(v|t_{k+1}) - \bar{x}^0(v|t_k)\| \|\bar{x}^0(v|t_k)\| dv \\ &\quad + \bar{\lambda}(Q) \int_{t_{k+1}}^{t_{k+1}+\mathcal{T}} \|\tilde{x}(v|t_{k+1}) - \bar{x}^0(v|t_k)\|^2 dv \\ &\leq \max_{v \in \mathbf{R}_{[t_{k+1}, t_k+\mathcal{T}]}} \|\tilde{x}(v|t_{k+1}) - \bar{x}^0(v|t_k)\| \\ &\quad \times \bar{\lambda}(Q) \int_{t_{k+1}}^{t_{k+1}+\mathcal{T}} \|\tilde{x}(v|t_{k+1}) - \bar{x}^0(v|t_k)\| dv \\ &\quad + \max_{v \in \mathbf{R}_{[t_{k+1}, t_k+\mathcal{T}]}} \|\bar{x}^0(v|t_k)\| \\ &\quad \times 2\bar{\lambda}(Q) \int_{t_{k+1}}^{t_{k+1}+\mathcal{T}} \|\tilde{x}(v|t_{k+1}) - \bar{x}^0(v|t_k)\| dv. \end{aligned}$$

According to (10) and (16), it is seen that

$$\mathcal{J}_{\Delta 1}^{0,k} \leq \bar{\lambda}(Q)(1 - \beta)\mathcal{T}\varpi e^{\ell(\mathcal{T}-\beta\mathcal{T})}(\varpi e^{\ell(\mathcal{T}-\beta\mathcal{T})} + 2\bar{\psi}),$$

where $\bar{\psi} = \sup_{x \in \mathcal{X}} \|x\| - \zeta(\beta\mathcal{T})$.

For (20b), by using (16) again, we have

$$\begin{aligned} \mathcal{J}_{\Delta 2}^{0,k} &\leq (\mathcal{V}_P^{\frac{1}{2}}(\tilde{x}(t_k + \mathcal{T}|t_{k+1})) + \mathcal{V}_P^{\frac{1}{2}}(\bar{x}^0(t_k + \mathcal{T}|t_k))) \\ &\quad \times \mathcal{V}_P^{\frac{1}{2}}(\tilde{x}(t_k + \mathcal{T}|t_{k+1}) - \bar{x}^0(t_k + \mathcal{T}|t_k)) \\ &\leq \sqrt{\bar{\lambda}(P)\varpi} e^{\ell(\mathcal{T}-\beta\mathcal{T})}(\gamma + \epsilon). \end{aligned}$$

For (20c), according to $x(t_k) \notin \Omega(\gamma)$, it is seen that

$$\mathcal{J}_{\Delta 3}^{0,k} \leq - \int_{t_k}^{t_{k+1}} \mathcal{V}_Q(\bar{x}^0(v|t_k)) dv \leq -\beta\mathcal{T}\bar{\lambda}(Q)\left(\frac{\gamma}{\sqrt{\bar{\lambda}(P)}} - \varpi\right)^2.$$

According to (17) and using the same statements as in Michalska and Mayne (1993), we have that $x(t)$ will get into the set $\Omega(\gamma)$ within a finite period.

Then, for $x(t_k) \in \Omega(\gamma)$, the control input $Kx(t)$ is enabled. Consider the Lyapunov function $\mathcal{V}_P(x(t))$. According to Lemma 1, it is seen that $\dot{\mathcal{V}}_P(x(t)) \leq -\mathcal{V}_Q^*(x(t)) + 2x^T(t)P\zeta(t) \leq \frac{\lambda(Q^*)}{\lambda(P)}(-\mathcal{V}_P(x(t)) + \bar{\gamma}^2)$. Then, by using the same statements in Li and Shi (2014), it is shown that, if $\bar{\gamma} \leq \gamma$, the closed-loop system will remain stable in $\Omega(\gamma)$ and $x(t)$ will converge to the set $\Omega(\bar{\gamma})$.

By summarizing these two cases, the proof is completed.

Remark 5. Theorems 2 and 3 imply that the feasibility and stability are related to the parameters $\mathcal{T}$, $\bar{\xi}_m$, and β . Generally, a larger β would lead to lower triggering frequency. However, less frequent sampling actions would lead to a decline in the control performance. Moreover, a larger $\mathcal{T}$ would result in improved control performance, as it enables consideration of the longer-term state evolution, which, however, would lead to increased computational resource costs. In accordance with Remark 3, more practical consideration in the parameter design of the DEMPC scheme can be directly inferred from the relevant literature on conventional EMPC schemes (Li & Shi, 2014; Liu et al., 2018; Sun et al., 2018).

5. Simulation study

The effectiveness of the proposed DEMPC algorithm is validated through two sets of simulations conducted on a cart-damper-spring system using the MATLAB platform, performed on a laptop with a 2.21 GHz processor and 8 GB of RAM. The dynamic model has the same form as in Rubagotti, Raimondo, Ferrara, and Magni (2011) and is expressed as

$$\dot{x}_1(t) = x_2(t) + p_1(t)$$

$$\dot{x}_2(t) = \frac{1}{M}(s(x(t))x_1(t) - h_d x_2(t) + u(t) + p_2(t)),$$

where $x_1(t)$ denotes the displacement of the cart from the equilibrium position, and $x_2(t)$ represents its velocity. $u(t)$ is the control input applied to the cart. $M = 1$ kg and $h_d = 1.1$ N s/m are the mass of the cart and the damper factor, respectively. $s(x(t)) = -k_0 e^{-x_1(t)}$ is the stiffness of the spring with $k_0 = 0.33$ N/m. The state and control input constraints herein are given as follows: $|x_1(t)| \leq 1$ m, $|x_2(t)| \leq 2$ m/s, and $|u(t)| \leq 2$ N, for $\forall t \in \mathbf{R}_{[0;\infty)}$. $p_1(t)$ and $p_2(t)$ are the unmatched and matched disturbances, respectively. The constant ℓ is derived as 2.89. The sampling interval is set to 0.05 s, while the prediction horizon $\mathcal{T}$ is set to 1 s. The initial state of the cart-damper-spring system is given as $x_0 = [0.95; -0.05]$. The simulation time is set to 20 s.

Case 1: Full Disturbances: Consider $p_1(t) = p_2(t) = 0.0042 \sin(3t + 1) + 0.0036 \cos(t)$, $\forall t \in \mathbf{R}_{[0;\infty)}$, thus $\rho_1 = 0.0077$ and $\rho_2 = 0.0155$. The observer parameter is designed as $\mathcal{L} = 100\mathcal{D}^+ = [50, 50]$ where $\mathcal{D} = [1; 1]$, following Corollary 1 in Xie et al. (2022). The parameter β in the triggering condition (11) is set to 0.9. The weighting matrixes are chosen as $Q = [23, 0; 0, 23]$ and $R = 4$. According to Lemma 1 and Theorems 2 and 3, we obtain that $\gamma = 2.50$, $\epsilon = 2.12$, $K = [2.09, 2.23]$, and $P = [30.86, 8.36; 8.36, 8.95]$. The DEMPC algorithm is implemented upon the above observer and MPC parameters and is in comparison with the DMPC algorithm (Xie et al., 2022). By following Theorem 2 in Xie et al. (2022), the parameters of the DMPC algorithm are configured as follows: $\mathcal{L} = [50, 50]$, $Q = \text{diag}\{23, 23\}$, $R = 6$, $\gamma = 3.60$, $\epsilon = 3.54$, $K = [1.65, 1.79]$, and $P = [32.25, 9.93; 9.93, 10.74]$. The displacement and velocity trajectories under the above two control algorithms are illustrated in Fig. 1. The control inputs and the corresponding execution time are reported in Fig. 2. Table 1 shows the triggering times (TT) of the algorithms. It can be seen from Table 1 and Fig. 2 that both the TT and average execution time associated with the DEMPC algorithm would be notably lesser compared to the

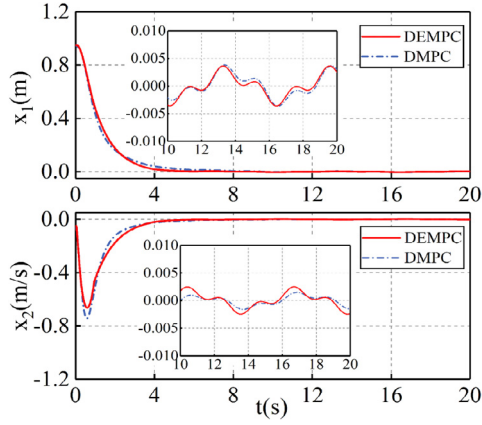


Fig. 1. Displacements and velocities under DMPC and DEMPC algorithms.

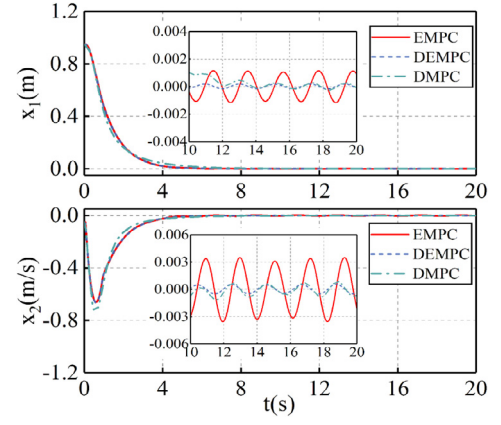


Fig. 3. Displacements and velocities under DMPC, EMPC, and DEMPC algorithms.

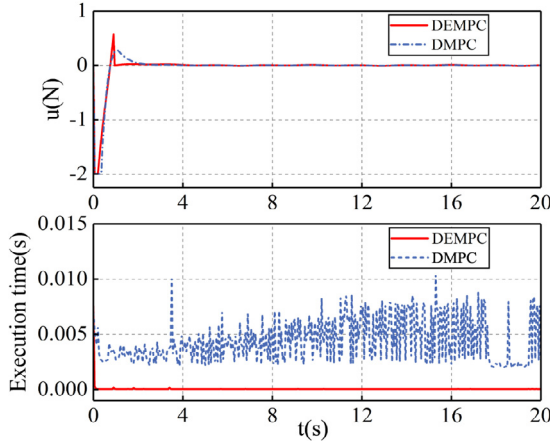


Fig. 2. Control inputs and execution time under DMPC and DEMPC algorithms.

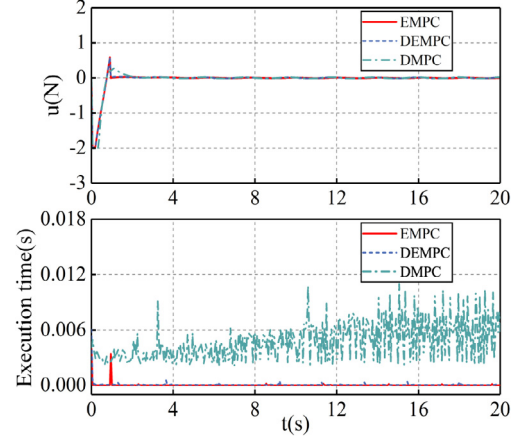


Fig. 4. Control inputs and execution time under DMPC, EMPC, and DEMPC algorithms.

Table 1

Comparison of TT of algorithms in two cases.

Algorithms (★:MPC)	Case 1		Case 2		
	D★	DE★	D★	E★	DE★
TT	400	1	400	2	1

DMPC algorithm because of the CTETM and dual-mode strategy, while the steady-state performance and transient performance of both algorithms remain comparable within the specified parameter environments. Therefore, the DEMPC algorithm could save more computational resources without significant compromise on control performance compared to the DMPC algorithm.

Case 2: Only Matched Disturbances: Consider $p_1(t) = 0$, $p_2(t) = 0.012 \sin(3t + 1) + 0.001 \cos(4t)$, $\forall t > 0$, thus $\rho_1 = 0.0127$ and $\rho_2 = 0.0392$. The observer gain is designed as $\mathcal{L} = 100\mathcal{D}^+ = [0, 100]$ where $\mathcal{D} = [0; 1]$. The DEMPC algorithm is implemented based on the following MPC parameters: $\beta = 0.9$, $Q = [23, 0; 0, 23]$, $R = 5$, $\gamma = 2.40$, $\epsilon = 1.98$, $K = [1.84, 1.98]$, and $P = [31.61, 9.20; 9.20, 9.90]$. In this case, the DEMPC algorithm is in comparison with the EMPC and DMPC algorithms to show the superiority as asserted in Remark 2. The EMPC algorithm is implemented by setting $u_{d0} = -B^+ \mathcal{D} \hat{p}(t) = 0$ with $\mathcal{U} = \mathcal{U}$, and is adapted to the parameter configuration of the DEMPC algorithm. The parameters of the DMPC algorithm are chosen as follows: $\mathcal{L} = [0, 100]$, $Q = [20, 0; 0, 20]$, $R = 6$, $\gamma = 3.90$, $\epsilon = 3.84$, $K = [1.52, 1.65]$, and $P = [28.49, 9.15; 9.15, 9.93]$. The corresponding results are

shown in Figs. 3–4 and Table 1. It is worth mentioning that, aside from reaffirming the analogous conclusions reached regarding the DEMPC and DMPC algorithms as in Case 1, the DEMPC algorithm, as analyzed in Remark 2, would outperform the EMPC algorithm in both control performance and computational resource usage in the specified parameter environment. Consequently, the DEMPC algorithm would achieve a more favorable trade-off between control performance and computational resources than the other two algorithms, which would be attributed to the harmonious integration of EMPC and disturbance observer technique.

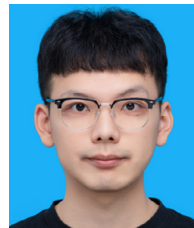
6. Conclusion

A novel DEMPC framework is proposed for a class of nonlinear input-affine continuous-time systems with entire state and control input constraints and unmatched disturbances. The primary aim of this framework is to save computational resources and improve the robustness of standard MPC simultaneously. In contrast to the conventional DMPC and EMPC algorithms, the DEMPC algorithm would attain a more favorable trade-off between computational resources and control performance. This advantage is particularly pronounced for systems with only matched disturbances. The theoretical properties involving robust constraint fulfillment, Zeno phenomenon prevention, recursive feasibility, and stability are investigated. Furthermore, two sets of comparative simulations are conducted to show the superiorities of the

DEMPC algorithm. Future research endeavors will focus on investigating time-varying DEMPC schemes aimed at further reducing constraint conservatism.

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